MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

BEAKER, EXPLOSIVE, 5"/54 PROJECTILE

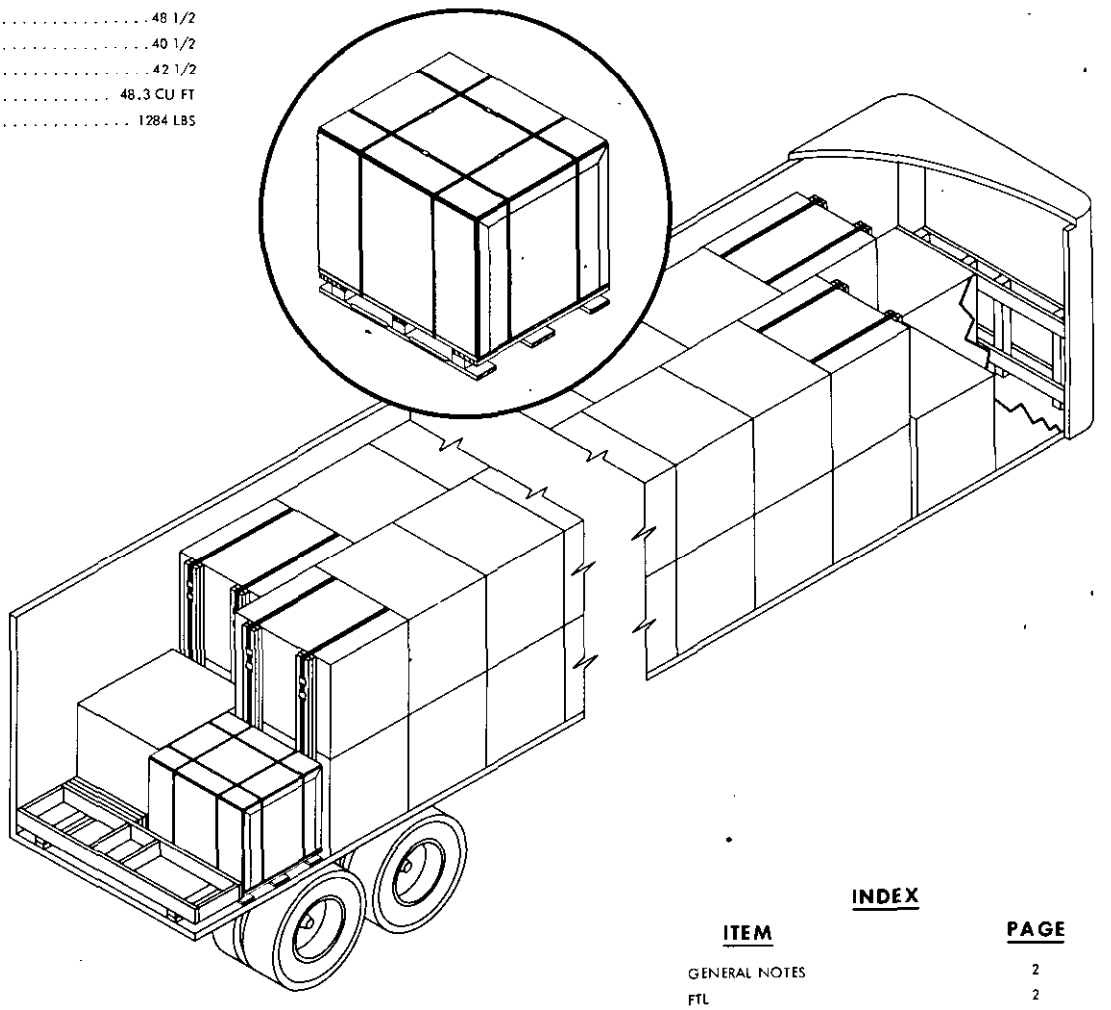
DOMESTIC UNIT LOAD

MIL-STD-1320-161
(NAVY)

30 SEPTEMBER 1975

UNIT LOAD DATA

UNIT LOAD DWG	WR-53/863
LENGTH	48 1/2
WIDTH	40 1/2
HEIGHT	42 1/2
CUBE	48.3 CU FT
WEIGHT (APPROX)	1284 LBS



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS. REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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FSC 8140

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N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING DOMESTIC UNIT LOADS OF THE EXPLOSIVE BEAKER FOR THE 5" / 54 PROJECTILE WHEN PALLETIZED IN ACCORDANCE WITH WR-53/863.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT TRAILERS WITH WOOD FLOORS, METAL FLOORS AND METAL FLOORS WITH WOOD NAILING STRIPS. TRAILERS SHALL HAVE THE TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. TO FIND THE CORRECT LOAD PATTERN TO USE:
   a. FIND THE LOAD PATTERN NUMBER IN TABLE I USING THE NUMBER OF UNIT LOADS TO BE LOADED INTO THE TRAILER AND THE LENGTH OF THE TRAILER TO BE LOADED.
   b. USING THE LOAD PATTERN NUMBER, ENTER TABLE II AND FIND THE LOAD PATTERN TO BE USED.
   c. LOAD, BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 5 AND 6.
4. THE CORRECT REAR BLOCKING TO BE USED DEPENDS UPON THE DISTANCE BETWEEN THE REAR OF THE LADING AND THE TRAILER DOORS WHEN CLOSED. WHEN THAT DISTANCE :

   LESS THAN 12 INCHES

   INSTALL SOLID FILL TO BEAR AGAINST LADING AND TRAILER DOORS. IF THE LADING IS NOT EVEN ACROSS THE REAR OF TRAILER, SHIM AS REQUIRED TO FILL VOID BETWEEN SOLID FILL AND LADING. SEE DETAIL A.

   WARNING

   REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

   12 TO 36 INCHES

   INSTALL REAR BLOCKING ASSEMBLY, DETAIL B, PAGE 2, TO BEAR AGAINST LADING AND TRAILER DOORS (SEE PAGE 5). IF LADING IS NOT EVEN ACROSS REAR OF TRAILER, SHIM AS REQUIRED TO FILL VOID BETWEEN ASSEMBLY AND LADING.

   WARNING

   REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

   GREATER THAN 36 INCHES

   INSTALL REAR BLOCKING AS SHOWN ON PAGE 6. SINCE BLOCKING MUST BE NAILED TO TRAILER FLOOR, DO NOT USE TRAILERS WITH METAL FLOORS.
5. TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND LAYER, THE UNIT LOADS SHALL BE UNITIZED (AS SHOWN ON PAGE 6) WHERE THE LAYERS CHANGE FROM TWO LAYERS TO ONE LAYER HIGH. THE STIFFENERS SHALL BE POSITIONED ON THE UNIT LOADS SO THAT WHEN THE UNIT LOADS ARE IN THE TRAILER, THE STRAPPING IS PARALLEL TO THE LONGITUDINAL AXIS OF THE TRAILER. THE STIFFENERS SHALL BE POSITIONED TOWARDS THE UNIT LOADS THAT ARE ONE HIGH (SEE PAGE 5). IF BOTH ENDS OF THE STACKED LOADS ARE AGAINST ONE HIGH UNIT LOADS, STIFFENERS SHALL BE USED ON BOTH ENDS OF THE STACKED TWO HIGH LOADS.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH EXPLOSIVES A PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40,42,44 & 45 FT TRAILERS (VAN)

1. A FULL TRUCKLOAD CONSISTS OF 35 UNIT LOADS WHEN LOADED IN A LIGHT WEIGHT TRACTOR-TRAILER (NET VEHICLE WEIGHT 26,000 LBS MAXIMUM) OR 33 UNIT LOADS WHEN LOADED IN A STANDARD WEIGHT TRACTOR-TRAILER (NET VEHICLE WEIGHT 27,500 LBS MAXIMUM).

WARNING

ONLY LIGHT WEIGHT TRACTOR-TRAILERS CAN BE LOADED WITH MORE THAN 33 UNIT LOADS. BEFORE ATTEMPTING TO LOAD 34 OR 35 UNIT LOADS, THE TARE WEIGHT OF THE VEHICLE SHOULD BE DETERMINED. IF THIS TARE WEIGHT IS OVER 26,000 LBS ONLY 33 OR LESS UNIT LOADS SHALL BE LOADED.

LTL 40,42,44 & 45 FT TRAILERS (VAN)

1. WHEN REQUIRED TO SHIP LESS THAN FTL DETERMINE LOAD PATTERN TO BE USED AS DESCRIBED IN GENERAL NOTE 3. BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 5 AND 6. REAR BLOCKING SHALL BE IN ACCORDANCE WITH GENERAL NOTE 4.

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*TW = TRAILER WIDTH

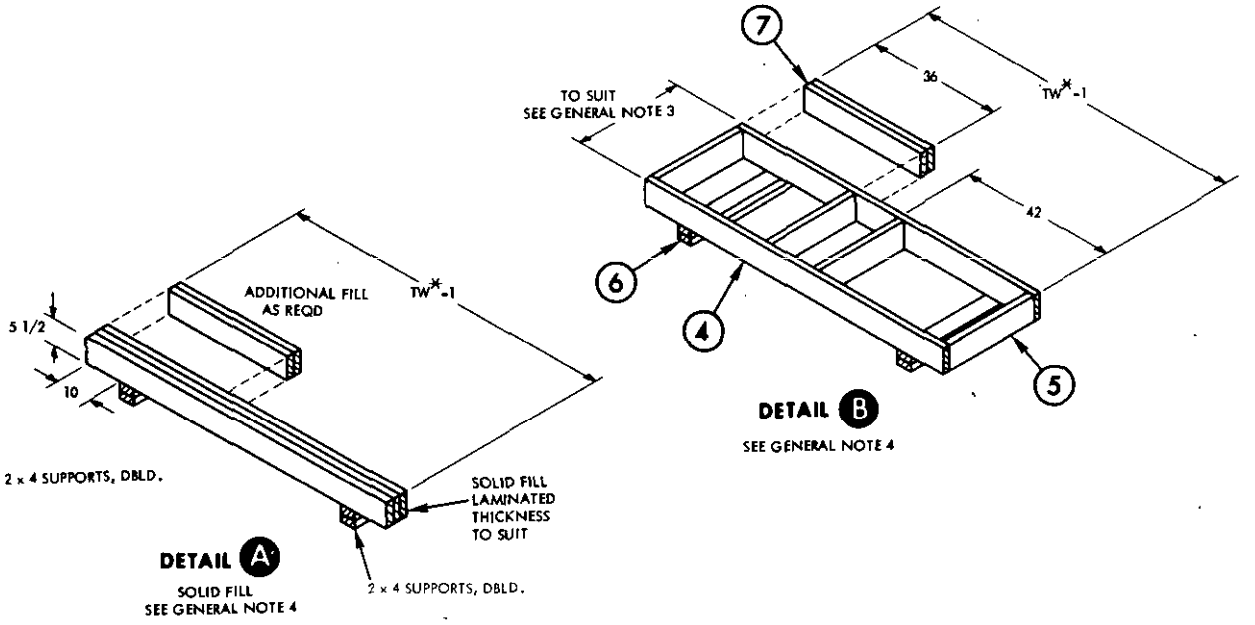


TABLE I

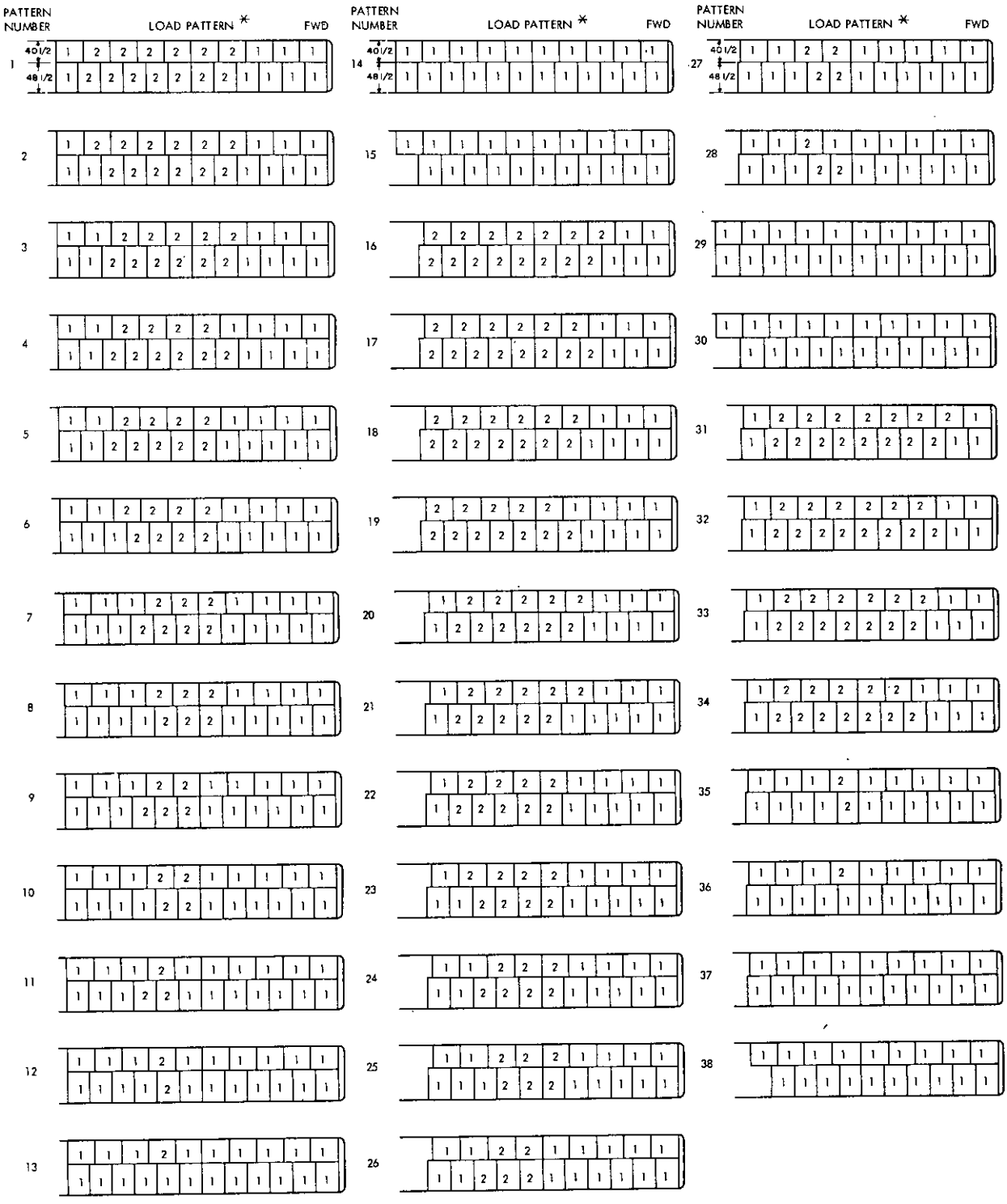
NO. OF UNIT LOADS TO BE SHIPPED	FTL OR LTL	LOAD PATTERN NO. TRAILER LENGTH (FT) 40	42	44/45	TRUCKLOAD WEIGHT (LBS) APPROX.
35	FTL*	31	16	1	44,940
34	LTL*	32	17	2	43,656
33	FTL/LTL	33	18	3	42,372
32	LTL	34	19	4	41,088
31	LTL	20	20	5	39,804
30	LTL	21	21	6	38,520
29	LTL	22	22	7	37,236
28	LTL	23	23	8	35,952
27	LTL	24	24	9	34,668
26	LTL	25	25	10	33,384
25	LTL	26	26	11	32,100
24	LTL	27	27	12	30,816
23	LTL	28	28	13	29,532
22	LTL	35	29	14	28,248
21	LTL	36	30	15	26,964
20	LTL	37	37	37	25,680
19	LTL	38	38	38	24,369
18 OR LESS	LTL	SEE NOTE AT BOTTOM OF TABLE II			

* **WARNING**

THIS LOAD IS SUITABLE FOR LIGHTWEIGHT TRACTOR-TRAILERS ONLY. IF LOADED ON A STANDARD WEIGHT TRACTOR TRAILER, THE AXLE WEIGHTS AND THE GROSS VEHICLE WEIGHT MAY BE IN EXCESS OF THE LEGAL LIMITS OF 32,000 LBS FOR TANDEM AXLES AND 73,280 LBS GROSS VEHICLE WEIGHT.

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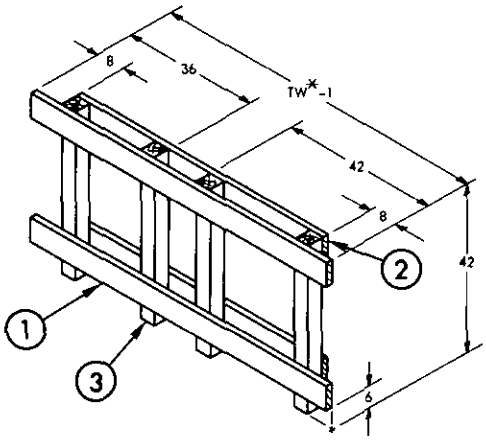
TABLE II



* LOAD PATTERN SHOWS TRAILER FLOOR PLAN. NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN A LAYER.

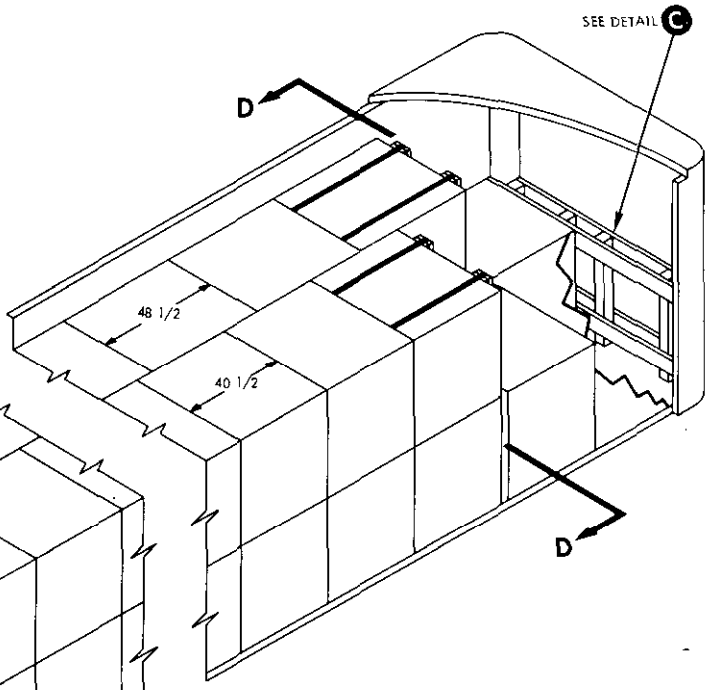
18 OR LESS UNIT LOADS:
POSITION UNIT LOADS IN THE TRAILER USING A DOUBLE ROW PATTERN. ONE ROW WITH THE 40 1/2 DIMENSION ACROSS THE TRAILER AND THE OTHER ROW THE 48 1/2 DIMENSION ACROSS THE TRAILER.

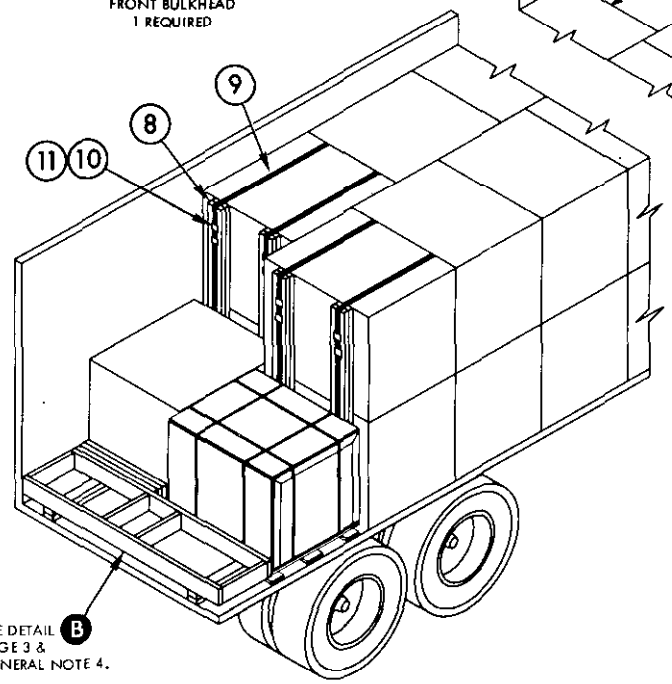
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DETAIL C

FRONT BULKHEAD
1 REQUIRED





SEE DETAIL B
PAGE 3 &
GENERAL NOTE 4.

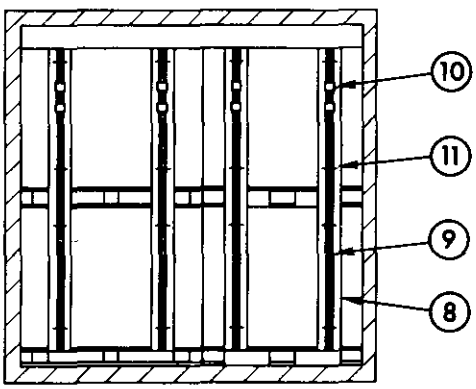
NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS IS 12 TO 36 INCHES. SEE GENERAL NOTE 4 FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER CONFIGURATIONS.

TRUCKLOAD DATA

SEE TABLE I

*TW = TRAILER WIDTH

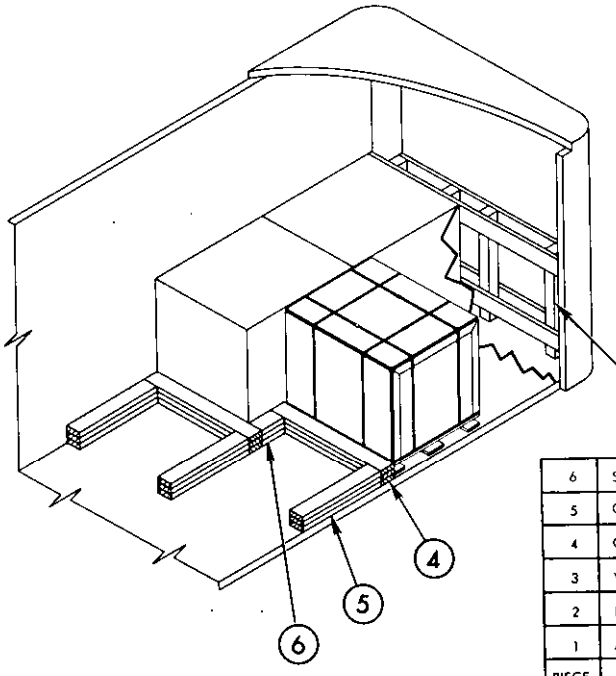


SECTION D-D

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
LIST OF MATERIALS AND NAILING DATA						
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	3	3	12d
2	FWD HORIZONTAL	2 x 6 TO SUIT	2	3	3	12d
3	VERTICAL	4 x 4 x 42	4	SEE 1 & 2	-	-
4	CROSSMEMBER	2 x 6 x TW*-1	2	5	3/JOINT	16d
5	STRUT	2 x 6 TO SUIT	4	SEE 4	-	-
6	SUPPORT	2 x 4 TO SUIT	4	4 6	2/JOINT 3	10d 10d
7	SHIM	6 WIDE, THICKNESS & LENGTH TO SUIT	AS REQD	4	AS REQD	12d
8	STIFFNER	2 x 6 x 80	16	DOUBLE LAMINATE	5	10d
9	STRAP	1 1/4 x .035 x 23 1/2 FT	8	-	-	-
10	SEAL	FOR 1 1/4 STRAP	16	-	-	-
11	STAPLE	FOR 1 1/4 STRAP	32	-	-	-

TYPICAL FULL BOTTOM LAYER & PARTIAL SECOND LAYER

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NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS IS GREATER THAN 36 INCHES AND THE TRAILER FLOOR IS METAL WITH WOOD NAILING STRIPS. WOOD FLOOR TRAILERS MAY ALSO BE USED.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
6	SPACER CLEAT	2 x 6 TO SUIT	3	2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.		
5	CLEAT	2 x 6 x 36	9			
4	CROSSMEMBER	2 x 6 x 48	6			
3	VERTICAL	4 x 4 x 42	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 x 6 TO SUIT	2	3	3	12d
1	AFT HORIZONTAL	2 x 6 x TW* -1	2	3	3	12d

LIST OF MATERIALS AND NAILING DATA

TYPICAL PARTIAL SINGLE LAYER

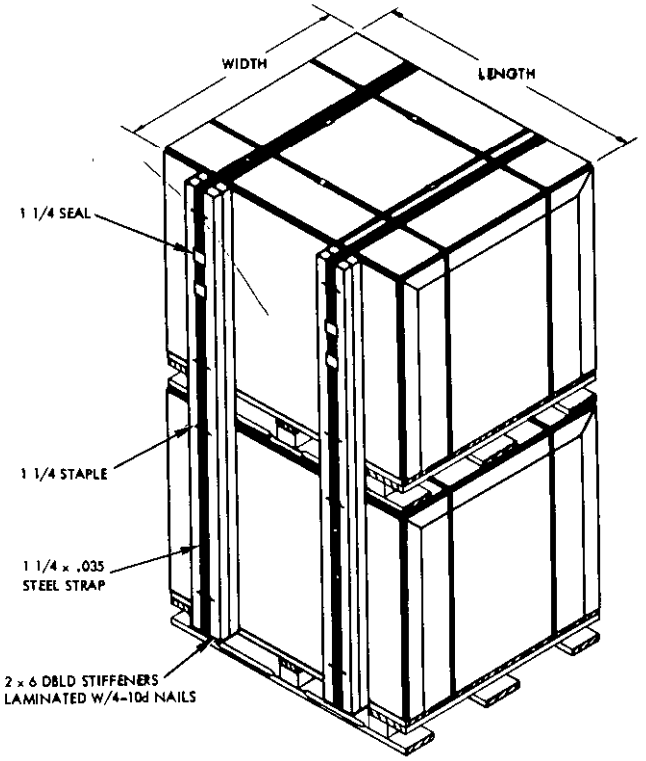
UNITIZING PROCEDURE

1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD, STACKED UNIT LOADS SHALL BE UNITIZED.

WARNING

THE LOAD PATTERN REQUIRES UNITIZING IN TWO CONFIGURATIONS: WITH THE STIFFENERS ON THE "LENGTH" SIDE OF THE STACKED UNIT LOADS AS SHOWN ABOVE AND STIFFENERS ON THE "WIDTH" SIDE. UNITIZING SHALL BE ACCOMPLISHED SO THAT THE REQUIREMENTS OF GENERAL NOTE 5 ARE SATISFIED.

2. THE DOUBLED 2 x 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE PALLET OF THE BOTTOM UNIT LOAD.
3. THE 1 1/4 INCH STEEL STRAPS POSITIONED AS SHOWN, ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE TOP DECK OF THE BOTTOM PALLET.
4. TENSION STRAPS AND SEAL WITH TWO CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STRAP STAPLES.



REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N295)

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